

Equivalenze logiche

Mostrare che le seguenti formule sono FO-equivalenti, usando le equivalenze logiche notevoli:

$$\exists x (P(x) \rightarrow \forall y Q(y)) \Leftrightarrow \exists x \forall y (P(x) \rightarrow Q(y)) \Leftrightarrow \forall y \exists x (P(x) \rightarrow Q(y))$$

$$\forall x (P(x) \rightarrow \neg Q(x)) \Leftrightarrow \neg \exists x (P(x) \wedge Q(x))$$

$$\forall x P(x) \rightarrow \exists y Q(y) \Leftrightarrow \exists x \exists y (P(x) \rightarrow Q(y))$$

Equivalenze logiche

$$\exists x (P(x) \rightarrow \forall y Q(y))$$

$$\Leftrightarrow$$

$$\exists x (\neg P(x) \vee \forall y Q(y)) \quad ; \text{rimpiazzamento, implicazione materiale}$$

$$\Leftrightarrow$$

$$\exists x \forall y (\neg P(x) \vee Q(y)) \quad ; \text{spostamento: } \forall z(A \vee B(z)) \Leftrightarrow A \vee \forall zB(z)$$

; z non occorre libera in A

$$\Leftrightarrow$$

$$\exists x \forall y (P(x) \rightarrow Q(y)) \quad ; \text{rimpiazzamento, implicazione materiale}$$

Equivalenze logiche

$$\exists x (P(x) \rightarrow \forall y Q(y))$$

$$\Leftrightarrow$$

$$\exists x (\neg P(x) \vee \forall y Q(y)) \quad ; \text{rimpiazzamento, implicazione materiale}$$

$$\Leftrightarrow$$

$$\exists x \neg P(x) \vee \forall y Q(y) \quad ; \text{spostamento: } \exists z(A(z) \vee B) \Leftrightarrow \exists zA(z) \vee B$$

$$\Leftrightarrow$$

$$\forall y (\exists x \neg P(x) \vee Q(y)) \quad ; \forall z(A \vee B(z)) \Leftrightarrow A \vee \forall zB(z), z \text{ non occorre libera in } A$$

$$\Leftrightarrow$$

$$\forall y \exists x (\neg P(x) \vee Q(y)) \quad ; \text{rimpiazzamento;} \\ ; \text{spostamento: } \exists z(A(z) \vee B) \Leftrightarrow \exists zA(z) \vee B$$

$$\Leftrightarrow$$

$$\forall y \exists x (P(x) \rightarrow Q(y)) \quad ; \text{rimpiazzamento, implicazione materiale}$$

Equivalenze logiche

$$\forall x (P(x) \rightarrow \neg Q(x))$$

$$\Leftrightarrow$$

$$\forall x (\neg P(x) \vee \neg Q(x)) \quad ; \text{rimpiazzamento, implicazione materiale}$$

$$\Leftrightarrow$$

$$\forall x \neg(P(x) \wedge Q(x)) \quad ; \text{rimpiazzamento, De Morgan}$$

$$\Leftrightarrow$$

$$\neg \exists x (P(x) \wedge Q(x)) \quad ; \text{De Morgan sui quantificatori}$$

Equivalenze logiche

$$\forall x P(x) \rightarrow \exists y Q(y)$$

$$\Leftrightarrow$$

$$\neg \forall x P(x) \vee \exists y Q(y) \quad ; \text{ implicazione materiale}$$

$$\Leftrightarrow$$

$$\exists x \neg P(x) \vee \exists y Q(y) \quad ; \text{ De Morgan sui quantificatori}$$

$$\Leftrightarrow$$

$$\exists x \exists y (\neg P(x) \vee Q(y)) \quad ; \text{ spostamento: } \exists z (A(z) \vee B) \Leftrightarrow \exists z A(z) \vee B$$

; rimpiazzamento,

$$; \text{ spostamento: } \exists z (A \vee B(z)) \Leftrightarrow A \vee \exists z B(z)$$

$$\Leftrightarrow$$

$$\exists x \exists y (P(x) \rightarrow Q(y)) \quad ; \text{ rimpiazzamento, implicazione materiale}$$